

GFNO-PINO: A Topology-Disjoint Benchmark Isolating Physics Fine-Tuning, Topology Conditioning, and Spectral Filtering in AC-OPF Graph Surrogates

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Abstract

Security assessment requires repeated alternating-current optimal power-flow (AC-OPF) solutions as injections and line availability change. Although topology inputs and spectral graph filters are architecturally plausible for this task, their benefit relative to physics-informed fine-tuning has not been isolated under matched, topology-disjoint comparisons. To test these choices, we construct a topology-conditioned Chebyshev graph surrogate (GFNO-PINO) that maps nodal injections, directional branch admittances, line status, equipment limits, and generator economics to voltage phasors and dispatch. It combines a supervised PowerModels-IPOPT warm start with exact differentiable AC residuals and zero-safe gradient-norm balancing; independent voltage and generator boxes and the reference angle are enforced by construction, while coupled AC constraints remain soft. The five-seed pilot benchmark covers IEEE 30-, 57-, 118-, and 300-bus cases, three unseen line-outage topologies per case, and five neural models. Relative to the data-only GFNO, paired physics fine-tuning reduces mean sample-maximum power-balance mismatch by 63.4–89.3%, and all 20 paired seed effects are positive; the accompanying seed-resampling intervals are descriptive, not population-confidence claims. In contrast, the parameter-matched topology-blind GFNO and spatial GNN match or outperform GFNO-PINO on three of four systems, so this benchmark does not establish a causal accuracy benefit from either explicit topology input or Chebyshev filtering. Every learned model obtains 0% feasibility under the declared proxy; GFNO-PINO’s mean maximum mismatch remains 0.281, 1.684, 0.890, and 5.959 p.u. from 30 to 300 buses. GPU forward-pass medians yield raw device-specific latency ratios of 8.3–138.1 $\times$ relative to stored IPOPT CPU solve medians, excluding verification and restoration. The contribution is therefore a controlled ablation identifying physics fine-tuning as the validated driver under this protocol and documenting unconfirmed architectural hypotheses, not a production-ready or exactly feasible solver.

Keywords: AC-OPF, Physics-Informed Learning, N-1 Contingency, Chebyshev Graph Neural Network, Renewable Integration

1. Introduction

Power-system operation is being reshaped by variable renewable generation, electrification, and increasingly active demand. Solar and wind injections can change substantially within an operating interval, while load uncertainty alters both active- and reactive-power requirements. At the same time, the N-1 security criterion requires operators to assess whether the system remains within voltage, generation, and branch-flow limits after the loss of any credible component. These interacting sources of variability turn dispatch into a sequence of closely spaced, nonlinear, topology-dependent optimization problems rather than a single deterministic calculation. Reviews of optimal power flow (OPF) and security-constrained OPF emphasize both their operational importance and their continuing computational difficulty [1–4].

The alternating-current OPF (AC-OPF) formulation represents voltage magnitudes, phase angles, active and reactive generation, losses, transformer effects, and thermal constraints. This fidelity produces non-convex equality

and inequality constraints. Mature nonlinear solvers such as IPOPT apply sparse primal-dual interior-point methods and are indispensable for offline planning and high-quality reference solutions [5]. Nevertheless, a complete solve is still required for each loading condition and each network topology. Under large scenario ensembles or online contingency screening, solver initialization, factorization, and convergence variability can become a latency bottleneck. Convex relaxations can provide valuable bounds but introduce their own tightness and scalability tradeoffs [4].

The direct-current OPF (DC-OPF) approximation is faster because it neglects reactive power, assumes voltage magnitudes near one per unit, and linearizes angle differences. These assumptions make DC-OPF useful for screening and market studies, but they can produce material errors under high loading, resistive networks, voltage-sensitive conditions, or binding reactive-power limits, as documented in [6]. A real-time AC-OPF surrogate should be benchmarked against DC-OPF for speed without treating the linear approximation as an AC-feasible reference.

Learning-based OPF methods amortize the cost of optimization: an expensive solver generates training examples offline, after which a neural network provides rapid on-line predictions. DeepOPF and related methods demonstrate the potential of predict-and-reconstruct strategies, feasibility-aware training, and topology embeddings [7–9]. Other work combines supervised learning with Lagrangian penalties or worst-case verification [10, 11]. However, a finite-dimensional network trained on one fixed ordering can learn spurious correlations specific to that network. Even when admittance is an input, dense architectures do not directly encode the locality, permutation structure, or graph spectrum of an electrical network.

Physics-informed neural networks (PINNs) reduce reliance on labels by penalizing governing-equation residuals [12, 13]. In power systems, PINNs have been used to embed power-flow or optimality equations [14–16]. A standard PINN nevertheless represents a solution for a fixed physical instance or uses a fixed-dimensional parameterization. If its residual is formed with one admittance matrix, replacing a line changes the governing operator itself. Retraining a separate PINN for each contingency defeats the intended real-time advantage.

Neural operators provide a more suitable abstraction: they learn maps between function spaces rather than a single input-output vector relationship. DeepONet learns an operator through branch and trunk networks [17], while the Fourier neural operator (FNO) parameterizes non-local integral kernels in a spectral domain [18, 19]. The original FNO assumes a regular Euclidean grid and a shared Fourier basis, whereas transmission networks are sparse, irregular graphs. Graph spectral methods replace Euclidean harmonics with functions of the graph Laplacian [20–22]. Chebyshev polynomial filters are particularly attractive under contingencies: they apply the same learned spectral multiplier to a changing Laplacian without requiring a topology-specific eigendecomposition or eigenvector alignment.

This paper conducts a controlled, topology-disjoint ablation of physics fine-tuning, explicit topology conditioning, and Chebyshev spectral filtering for AC-OPF graph surrogates. GFNO-PINO is the experimental instrument for separating these effects, not a presupposed superior architecture. The principal contributions are:

- A differentiable AC physics objective using exact bus-admittance and branch-current operators, coupled with adaptive gradient-norm loss balancing and a supervised-to-physics curriculum. Hard transformations enforce independent voltage and generator boxes and the slack reference, while all coupled constraints that remain approximate are reported.
- A reproducible five-seed, topology-disjoint pilot benchmark on four IEEE systems against IPOPT AC-OPF, DC-OPF, a base-only PINN, a parameter-matched topology-blind GFNO, a spatial GNN, and a data-only graph FNO.

- A topology-conditioned graph-surrogate formulation, evaluated as a hypothesis under test, in which directional admittance, status, and rating alter both the learned propagation graph and the sample-specific AC residual.
- The validated empirical finding that paired physics fine-tuning reduces mean sample-maximum balance mismatch by 63.4–89.3% relative to data-only training under this pilot protocol, with all 20 paired seed effects positive.
- The negative empirical finding that explicit topology conditioning and Chebyshev filtering do not establish a causal accuracy benefit: matched topology-blind and spatial baselines match or outperform the full model on three of four systems. Every learned model remains 0% feasible under the declared proxy.

Reporting the unfavorable architectural ablations is part of the contribution: it distinguishes the validated effect (physics fine-tuning) from the unconfirmed effects (topology conditioning and spectral filtering) and informs which surrogate-design complexity is justified by the present evidence.

The remainder of the paper is organized as follows. Section 2 reviews classical, learning-based, physics-informed, and neural-operator approaches. Section 3 presents the AC-OPF formulation and the benchmarked GFNO-PINO architecture. Section 4 defines the benchmark protocol. Section 5 reports and discusses the executed benchmark. Section 6 states the scope of the feasibility and generalization claims, and Section 7 concludes the paper.

2. Related work

2.1. Classical and learning-based AC-OPF

AC-OPF is a foundational nonlinear optimization problem in power-system planning and operation. Standard formulations minimize generation cost subject to nonlinear nodal balances and equipment constraints [1, 2]. Sparse interior-point algorithms provide reliable local solutions and form the basis of widely used tools [5]. MATPOWER, pandapower, and PowerModels provide reproducible implementations and test-case interfaces [23–25]. Because AC-OPF is non-convex, a locally converged point is not in general a certificate of global optimality. Convex and moment relaxations may produce bounds or exact solutions in favorable cases, but no single relaxation dominates across all large practical networks [3, 4].

Supervised surrogates learn the mapping from uncertain demand to optimal decisions. DeepOPF predicts independent variables and reconstructs dependent quantities to improve feasibility [7, 8]. Fioretto et al. combine deep learning with a Lagrangian-dual objective [10]; Venzke et al. formulate optimization problems that bound worst-case neural-network violations [11]. These approaches show

that inference can be much faster than solving each instance from scratch, but feasibility treatment and generalization domains differ. DeepOPF-FT is especially relevant because it embeds topology and admittance to address multiple related AC-OPF problems [9]. The present work differs by using graph-spectral propagation that changes with the input topology and by training with exact AC residuals.

2.2. Physics-informed neural networks in power systems

PINNs augment data losses with residuals of governing equations [12]. The broader physics-informed machine-learning framework can improve data efficiency and physical consistency, but optimization can be stiff when residual terms have different units or gradient scales [13]. In power applications, Misyris et al. embedded power-system equations in neural training [14], and Lei et al. developed a physics-informed data-driven OPF framework [16]. Nellikkath and Chatzivasileiadis incorporated AC-OPF physics and methods for reducing worst-case violations [15].

Physics penalties do not automatically provide feasibility. A small mean residual can coexist with a large worst-bus error, and weighted penalties may trade economic performance against constraint satisfaction. Furthermore, a PINN residual evaluated using one fixed Y_{bus} does not describe the equations after a branch outage. The network may be fine-tuned for every new Y_{bus} , but such a workflow is not topology independent. The topology-conditioned formulation under test instead treats topology as part of the input function and evaluates every residual with the corresponding sample-specific admittance operators.

2.3. Neural operators and graph spectral methods

DeepONet establishes a practical architecture for learning nonlinear operators using separate encoders for input functions and output coordinates [17]. FNO learns a global convolution kernel in Fourier space and has demonstrated mesh-transfer properties for parametric partial differential equations [18]. Neural-operator theory formalizes discretization-aware mappings between infinite-dimensional spaces [19]. Applying a Euclidean FFT directly to buses, however, would assign physical meaning to an arbitrary bus ordering and would not respect irregular connectivity.

Graph signal processing defines spectral filters through the eigenstructure of the graph Laplacian [20]. Chebyshev networks approximate such filters by stable polynomials, avoiding explicit eigendecomposition [21]. Graph convolution and geometric deep learning extend local and spectral operations to non-Euclidean domains [22, 26]. In power systems, graph neural solvers exploit network structure for power flow and OPF [27, 28]. A spatial GNN aggregates nearby features, whereas a polynomial spectral filter parameterizes a frequency response of the topology-conditioned Laplacian. However, the implemented Chebyshev layer has the same localized polynomial form as

ChebNet. Because this study trains a separate checkpoint per system size and performs no mesh-refinement or discretization-transfer experiment, it does not claim the stronger function-space generalization normally associated with neural operators. “GFNO-PINO” is retained as the implementation name, not as evidence of such a property.

2.4. Research gap

The preceding literatures leave a specific gap. Classical methods account for topology and AC physics but solve every scenario separately. Supervised OPF surrogates provide low latency but can be tied to a fixed dimension and may require post-processing. PINNs inject physics but are commonly trained for a fixed admittance matrix. Neural operators target families of equations, yet standard FNOs assume a regular grid, and existing OPF GNNs do not necessarily combine graph-spectral propagation, sample-specific AC residuals, adaptive multi-objective training, and topology-disjoint evaluation. This work addresses that intersection. “Topology conditioned” means that one parameter set processes the base grid and unseen non-islanding line-status configurations within a fixed test-system family; it does not imply universal zero-shot transfer to arbitrary utilities, generator sets, or unbounded network sizes. Whether this input actually improves accuracy relative to simpler architectures is treated as an empirical question and tested directly in Section 5.5, where it is not confirmed under the present protocol.

3. Methodology

3.1. AC-OPF problem formulation

Let $\mathcal{N}$, $\mathcal{L}$, $\mathcal{X}$, $\mathcal{B} = \mathcal{L} \cup \mathcal{X}$, and $\mathcal{G}$ denote the bus, physical-line, fixed-transformer, full canonical-branch, and dispatchable-generator sets, respectively. Each branch $a \in \mathcal{B}$ has one canonical from/to orientation and two terminal equations. The physical-line status vector is $z \in \{0, 1\}^{|\mathcal{L}|}$, and its full-branch status map is

$$s_a(z) = \begin{cases} z_a, & a \in \mathcal{L}, \\ 1, & a \in \mathcal{X}. \end{cases} \quad (1)$$

Thus transformers participate in every graph and AC operator but are not outage candidates in this benchmark. Generator-to-bus incidence is represented by $\mathcal{G}_i \subseteq \mathcal{G}$. For bus i , the voltage phasor is $\mathcal{V}_i = V_i e^{j\theta_i}$. Active and reactive demand are P_i^d, Q_i^d , renewable active injection is P_i^r , and dispatchable generation is P_g^s, Q_g^s . With quadratic cost coefficients c_{2g}, c_{1g}, c_{0g} , AC-OPF is

$$\underset{V, \theta, P_g^s, Q_g^s}{\text{minimize}} \quad C(P^s) = \sum_{g \in \mathcal{G}} (c_{2g}(P_g^s)^2 + c_{1g}P_g^s + c_{0g}). \quad (2)$$

For topology z , let $Y(z) = G(z) + jB(z) \in \mathbb{C}^{|\mathcal{N}| \times |\mathcal{N}|}$ be the bus-admittance matrix. A base topology has $z_\ell = 1$ for every in-service line. An N-1 topology sets exactly one

eligible physical-line status to zero. With $z_k^{(\ell)} = 1 - \mathbb{I}[k = \ell]$, define the active full branch set $\mathcal{B}(z) = \{a \in \mathcal{B} : s_a(z) = 1\}$. The evaluated set is

$$\mathcal{T}_{N-1} = \{z^{(\ell)} \mid \ell \in \mathcal{L}_{\text{eligible}}, (\mathcal{N}, \mathcal{B}(z^{(\ell)})) \text{ connected}\}. \quad (3)$$

The active- and reactive-power injections implied by the voltage are

$$P_i(V, \theta; z) = V_i \sum_{j \in \mathcal{N}} V_j [G_{ij}(z) \cos(\theta_i - \theta_j) + B_{ij}(z) \sin(\theta_i - \theta_j)], \quad (4)$$

$$Q_i(V, \theta; z) = V_i \sum_{j \in \mathcal{N}} V_j [G_{ij}(z) \sin(\theta_i - \theta_j) - B_{ij}(z) \cos(\theta_i - \theta_j)]. \quad (5)$$

Kirchhoff's current law, written as nodal complex-power balance, requires

$$\sum_{g \in \mathcal{G}_i} P_g^g + P_i^r - P_i^d = P_i(V, \theta; z), \quad \forall i \in \mathcal{N}, \quad (6)$$

$$\sum_{g \in \mathcal{G}_i} Q_g^g - Q_i^d = Q_i(V, \theta; z), \quad \forall i \in \mathcal{N}. \quad (7)$$

Equations (4)–(7) are implemented in complex form as

$$S^{\text{net}} = \mathcal{V} \odot (Y(z)\mathcal{V})^*, \quad (8)$$

which is algebraically equivalent and avoids materializing a dense tensor of all angle differences.

Let $Y_f(z)$ and $Y_t(z)$ map bus voltages to branch currents at the from and to terminals. This representation retains line charging, off-nominal transformer ratios, and phase shifts. Terminal powers are

$$S_a^f = \mathcal{V}_{f(a)} ([Y_f(z)\mathcal{V}]_a)^*, \quad (9)$$

$$S_a^t = \mathcal{V}_{t(a)} ([Y_t(z)\mathcal{V}]_a)^*, \quad (10)$$

for $a \in \mathcal{B}$ with $s_a(z) = 1$. The remaining operating constraints are

$$P_g^{\min} \leq P_g^g \leq P_g^{\max}, \quad Q_g^{\min} \leq Q_g^g \leq Q_g^{\max}, \quad \forall g \in \mathcal{G}, \quad (11)$$

$$V_i^{\min} \leq V_i \leq V_i^{\max}, \quad \forall i \in \mathcal{N}, \quad (12)$$

$$\max\{|S_a^f|, |S_a^t|\} \leq S_a^{\max}, \quad \forall a \in \mathcal{B} : s_a(z) = 1, \quad (13)$$

$$\underline{\Delta\theta}_a \leq \theta_{f(a)} - \theta_{t(a)} \leq \overline{\Delta\theta}_a, \quad \forall a \in \mathcal{B} : s_a(z) = 1, \quad (14)$$

$$\theta_{i_{\text{ref}}} = 0. \quad (15)$$

A zero MATPOWER branch rating denotes an unconstrained line and is excluded from the thermal-violation denominator rather than interpreted as a zero-MVA limit.

3.2. Topology-conditioned surrogate formulation

For each bus, the input field is

$$x_i = [P_i^d, Q_i^d, P_i^r, \text{onehot}_4(\text{type}_i), V_i^{\text{set}}] \in \mathbb{R}^8. \quad (16)$$

The four bus categories are isolated, PQ, PV, and reference; the setpoint channel is zero at buses without a regulated generator. For each canonical branch $a = (i, j) \in \mathcal{B}$, the edge field is

$$e_a = [\Re Y_a^{ff}, \Im Y_a^{ff}, \Re Y_a^{ft}, \Im Y_a^{ft}, \Re Y_a^{tt}, \Im Y_a^{tt}, \Re Y_a^{tf}, \Im Y_a^{tf}, s_a(z), S_a^{\max}] \in \mathbb{R}^{10}. \quad (17)$$

where the four complex coefficients are the exact two-port terminal model. This directional representation retains tap ratios and phase shifts in the predictor as well as in the exact Y_f and Y_t physics operators.

The target solution field is $u = (V, \theta, P^g, Q^g)$. The learned finite-dimensional surrogate is

$$\mathcal{G}_{\Theta} : (X, E, z, \mathcal{M}_{\text{pred}}) \mapsto (\widehat{V}, \widehat{\theta}, \widehat{P}^g, \widehat{Q}^g), \quad (18)$$

where $\mathcal{M}_{\text{pred}}$ contains bus and generator box limits, generator-to-bus incidence, normalized quadratic cost coefficients, generator identity, and the slack-bus indicator. Branch angle limits are deliberately not predictor inputs: together with exact Y , Y_f , and Y_t , they belong to separate physics/evaluation metadata $\mathcal{M}_{\text{phys}}$. This separation matches the ten-channel edge field in Eq. (17). Parameters Θ are shared across every topology presented to one grid-specific checkpoint, but not across the four grid sizes.

3.3. Units and feature scaling

All nodal active/reactive quantities, generator limits and targets, branch MVA ratings, and admittance operators use the MATPOWER case base S_{base} and are stored in per unit. Voltage magnitude is per unit and phase variables and limits are radians. The bus field is not standardized with test-set statistics: its first three channels are physical per-unit values, the type channels are binary, and the setpoint is per unit. Directional terminal admittances are the per-unit coefficients returned by `makeY-bus`; status is binary and rating is $S_a^{\max}/S_{\text{base}}$. The physical cost is always evaluated after $P_g^{\text{MW}} = S_{\text{base}} P_g^{\text{p.u.}}$. For generator-head conditioning, (c_2, c_1, c_0) is first transformed to $(c_2 S_{\text{base}}^2, c_1 S_{\text{base}}, c_0)$ and divided by the case-level sum of absolute transformed coefficients (lower bounded by one). This makes the learned economic features dimensionless without changing the objective used for training or evaluation. Every result and table labels p.u., MW, Mvar, radians, currency/hour, and milliseconds explicitly.

3.4. Chebyshev graph-spectral architecture

For topology z , an undirected weighted adjacency matrix is assembled using

$$A_{ij}(z) = \sum_{a \in \mathcal{B} : \{f(a), t(a)\} = \{i, j\}} \frac{s_a(z)}{2} (|Y_a^{ft}| + |Y_a^{tf}|). \quad (19)$$

With $D = \text{diag}(\mathbf{A}\mathbf{1})$, the symmetric normalized Laplacian is

$$L(z) = I - D^{-1/2}A(z)D^{-1/2}. \quad (20)$$

Its spectrum lies in $[0, 2]$ for a nonnegative graph, so $\tilde{L} = L - I$ maps the spectrum to $[-1, 1]$ without computing a sample-specific largest eigenvalue.

A direct graph-Fourier layer would multiply in the eigenbasis $L = U\Lambda U^\top$. Under a line outage, eigenvectors can change sign, rotate within repeated eigenspaces, or exchange order. Consequently, coefficient channels do not have a consistent interpretation across topologies. We instead parameterize the spectral response by a K -term Chebyshev polynomial of degree $K - 1$:

$$\begin{aligned} T_0(\tilde{L})H &= H, & T_1(\tilde{L})H &= \tilde{L}H, \\ T_k(\tilde{L})H &= 2\tilde{L}T_{k-1}(\tilde{L})H - T_{k-2}(\tilde{L})H, & k &\geq 2, \end{aligned} \quad (21)$$

$$\mathcal{K}_\Theta^{(r)}(H; z) = \sum_{k=0}^{K-1} T_k(\tilde{L}(z))HW_k^{(r)}. \quad (22)$$

This is a graph-spectral multiplier, but it is basis-free and localized to $K - 1$ hops. It requires no eigendecomposition for a new contingency. Its localized polynomial form is also that of a Chebyshev graph neural network; the present benchmark does not demonstrate discretization invariance or an advantage over the spatial GNN.

Two endpoint-specific edge multilayer perceptrons encode e_a for the from and to buses; their outputs are multiplied by $s_a(z)$ so that a bias cannot leak through an outaged line. Endpoint-specific encodings are averaged using the active post-outage incident degree $\sum_{a \ni i} s_a(z)$, rather than the base degree, and concatenated with x_i . A linear lifting map produces width- d features $H^{(0)}$. Each GFNO block combines the polynomial spectral path with a learned pointwise path:

$$\tilde{H}^{(r+1)} = \text{GELU}\left(\mathcal{K}_\Theta^{(r)}(H^{(r)}; z) + H^{(r)}W_{\text{pw}}^{(r)} + b^{(r)}\right), \quad (23)$$

$$H^{(r+1)} = \text{LayerNorm}\left(H^{(r)} + \text{Dropout}(\tilde{H}^{(r+1)})\right). \quad (24)$$

Bus heads decode voltage magnitude and angle. Generator heads gather the latent feature at each generator bus and concatenate dimensionless limits, voltage setpoint, per-unit-transformed (c_2, c_1, c_0) , and normalized generator identity before decoding active and reactive dispatch. Thus, two units at one bus can have distinct economics and limits.

3.5. Physics-informed loss function

All powers in the network equations are in per unit on the case base S_{base} ; generator powers are converted to MW before evaluating the supplied cost polynomial. To prevent a soft-constrained model from being rewarded for reducing cost by shedding power, the implemented economic term is one-sided samplewise normalized regret,

$$\mathcal{L}_{\text{economic}} = \text{mean}_b \left[\text{ReLU} \left(\frac{C(\hat{P}_b^s) - C(P_b^{g,\text{ref}})}{\max(|C(P_b^{g,\text{ref}})|, C_{\text{scale}})} \right) \right]^2, \quad (25)$$

where $P_b^{g,\text{ref}}$ is the locally converged IPOPT dispatch and $C_{\text{scale}} = 1$ currency unit per hour. The ReLU is evaluated before the batch mean, so an over-cost sample cannot be cancelled by an under-cost sample; the formula also remains correct when a reference cost is negative. Lower predicted cost is not rewarded unless the other objectives retain the solver-informed dispatch and satisfy the AC equations. Let $r_{i,b}^P$ and $r_{i,b}^Q$ be the left side minus the right side of Eqs. (6) and (7). The equality residual is

$$\mathcal{L}_{\text{powerflow}} = \text{mean}_{(b,i) \text{ valid}} \left[(r_{i,b}^P)^2 + (r_{i,b}^Q)^2 \right]. \quad (26)$$

The two-ended thermal excess and penalty are

$$\begin{aligned} v_{a,b}^S &= \text{ReLU} \left(\max\{|\hat{S}_{a,b}^f|, |\hat{S}_{a,b}^t|\} - S_a^{\text{max}} \right), \\ \mathcal{L}_{\text{thermal}} &= \text{mean}_{(b,a) \in \mathcal{B}_S} (v_{a,b}^S)^2, \end{aligned} \quad (27)$$

where $\mathcal{B}_S$ contains only active, rated branches. The angle-difference excess is trained for every active branch:

$$\begin{aligned} v_{a,b}^\theta &= \text{ReLU} \left(\Delta\theta_a - (\hat{\theta}_{f(a),b} - \hat{\theta}_{t(a),b}) \right) \\ &\quad + \text{ReLU} \left((\hat{\theta}_{f(a),b} - \hat{\theta}_{t(a),b}) - \Delta\theta_a \right), \\ \mathcal{L}_{\text{angle}} &= \text{mean}_{(b,a):s_a(z_b)=1} (v_{a,b}^\theta)^2. \end{aligned} \quad (28)$$

Although voltage boxes are enforced by the output layer, their reported diagnostic is

$$\begin{aligned} v_{i,b}^V &= \text{ReLU}(V_i^{\text{min}} - \hat{V}_{i,b}) + \text{ReLU}(\hat{V}_{i,b} - V_i^{\text{max}}), \\ \mathcal{L}_{\text{voltage}} &= \text{mean}_{b,i} (v_{i,b}^V)^2. \end{aligned} \quad (29)$$

An analogous generator-box diagnostic is also logged. Neither identically zero hard-box diagnostic participates in adaptive loss balancing. Given solver labels u_b^{ref} , the warm-start term is

$$\mathcal{L}_{\text{data}} = \frac{1}{4} \sum_{q \in \{V, \theta, P^s, Q^s\}} \text{mean}_{(b,k) \text{ valid}} (\hat{q}_{b,k} - q_{b,k}^{\text{ref}})^2. \quad (30)$$

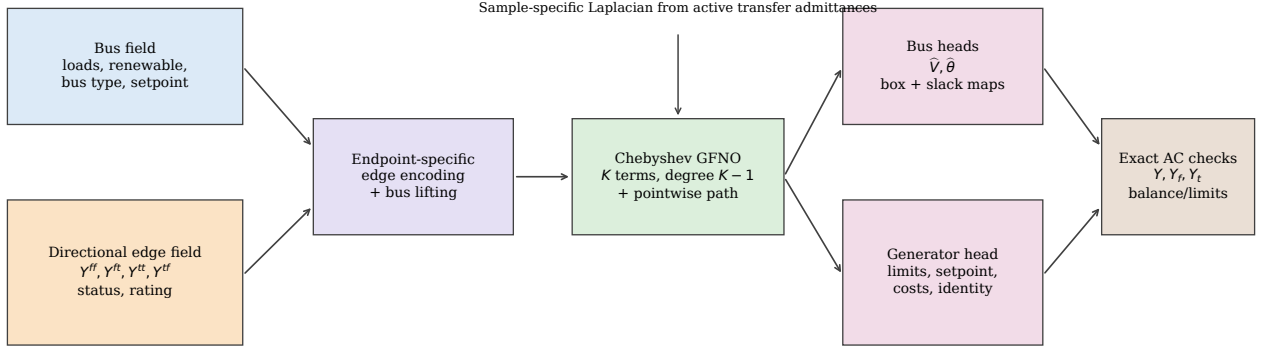
The total training objective is

$$\begin{aligned} \mathcal{L}_{\text{total}} &= \lambda_{\text{cost}} \mathcal{L}_{\text{economic}} + \lambda_{\text{pf}} s_{\text{pf}} \mathcal{L}_{\text{powerflow}} + \lambda_{\text{thermal}} s_{\text{th}} \mathcal{L}_{\text{thermal}} \\ &\quad + \lambda_{\text{angle}} \mathcal{L}_{\text{angle}} + \lambda_{\text{data}} \mathcal{L}_{\text{data}}. \end{aligned} \quad (31)$$

During physics epoch $e \in \{1, \dots, E_p\}$, define the exact objectives supplied to both the GradNorm update and the subsequent model update as

$$(\tilde{\mathcal{L}}_m)_m = (\mathcal{L}_{\text{economic}}, s_{\text{pf}}(e) \mathcal{L}_{\text{powerflow}}, s_{\text{th}}(e) \mathcal{L}_{\text{thermal}}, \mathcal{L}_{\text{angle}}, \mathcal{L}_{\text{data}}), \quad (32)$$

where $s_{\text{pf}}(e) = 1 + 9e/E_p$ and $s_{\text{th}}(e) = 1 + 4e/E_p$. Thus annealing changes the model gradient as well as the weight-balancing dynamics. To make GradNorm [29] well-defined when a constraint loss is initially zero, objective m enters the active set $\mathcal{A}$ only on the first step t_m for which its detached value exceeds $\varepsilon = 10^{-8}$; its baseline is then $\tilde{\mathcal{L}}_{m,0} = \tilde{\mathcal{L}}_m(t_m)$. Let $g_m = \|\nabla_{\Theta_s}(\lambda_m \tilde{\mathcal{L}}_m)\|_2$. Here $\Theta_s = W_{\text{lift}}$ is exactly



An outage changes inputs and exact physics operators; learned parameters are unchanged.

Figure 1: Topology-conditioned GFNO-PINO architecture. A line outage changes the edge status, the graph Laplacian, and the exact admittance operators supplied to the loss; learned parameters are unchanged.

the lift.weight matrix for GFNO and spatial-GNN models; the fixed-topology PINN uses network.0.weight. For the current active set, the positive weights are normalized as $\sum_{m \in \mathcal{A}} \lambda_m = |\mathcal{A}|$; inactive weights are exactly zero and do not enter this normalization. Let

$$\rho_m = \frac{(\text{stopgrad } \tilde{\mathcal{L}}_m + \varepsilon) / (\tilde{\mathcal{L}}_{m,0} + \varepsilon)}{\text{mean}_{j \in \mathcal{A}} [(\text{stopgrad } \tilde{\mathcal{L}}_j + \varepsilon) / (\tilde{\mathcal{L}}_{j,0} + \varepsilon)]}. \quad (33)$$

The weight update minimizes

$$\mathcal{L}_{\text{GN}} = \sum_{m \in \mathcal{A}} |g_m - \bar{g} \rho_m^\alpha|, \quad \bar{g} = \text{mean}_{m \in \mathcal{A}} g_m, \quad (34)$$

with $\alpha = 0.5$ and a stop-gradient target $\bar{g} \rho_m^\alpha$. If fewer than two objectives are active, the weight update is zero. The subsequent model update detaches the normalized weights; therefore the model and weight optimizers cannot differentiate through each other. Every raw loss, annealing factor, and effective weight is retained in the run artifacts; active-set transitions are deterministic from the archived losses and ε .

3.6. Hard-constraint enforcement layer

For unconstrained decoder values $a_i^V, a_g^P, a_g^Q \in \mathbb{R}$, the independent box constraints are enforced by

$$\widehat{V}_i = V_i^{\min} + (V_i^{\max} - V_i^{\min}) \sigma(a_i^V), \quad (35)$$

$$\widehat{P}_g^s = P_g^{\min} + (P_g^{\max} - P_g^{\min}) \sigma(a_g^P), \quad (36)$$

$$\widehat{Q}_g^s = Q_g^{\min} + (Q_g^{\max} - Q_g^{\min}) \sigma(a_g^Q), \quad (37)$$

where σ is the logistic sigmoid. For finite logits this map reaches the open interval, not its endpoints. It guarantees membership in the closed box but cannot represent a reference solution exactly at an active bound; this active-set limitation is quantified in Section 5. If $\tilde{\theta}$ is the raw angle vector, the reference is imposed by

$$\widehat{\theta}_i = \tilde{\theta}_i - \tilde{\theta}_{\text{ref}}. \quad (38)$$

Equations (35)–(38) guarantee the voltage and generator boxes and slack reference up to floating-point precision. They do not guarantee Eqs. (6), (7), (13), or (14); those coupled non-convex constraints remain approximate and must be measured after inference.

3.7. Training procedure

Training uses two phases. The first phase supplies a stable solver-informed initialization. The second phase jointly optimizes economic quality and physical consistency, with adaptive weights reducing sensitivity to disparate units and gradient scales. Topology IDs are sampled only from the training partition.

4. Case study and experimental setup

4.1. Test systems

Experiments are configured for the IEEE 30-, 57-, 118-, and 300-bus systems distributed through pandapower and compatible MATPOWER data [23, 24]. Table 1 reports structural facts extracted by the released data pipeline. “Eligible N-1” counts non-islanding outages of elements represented as lines in each pandapower case; transformer outages are not included in the present contingency set.

Every neural experiment uses seeds 2026–2030, Python 3.12.13, single-precision PyTorch 2.13.0 with Compute Unified Device Architecture (CUDA) 12.6, and an NVIDIA GeForce RTX 3050 6 GB Laptop graphics processing unit (GPU). Data handling uses pandapower 3.5.4, NumPy 2.4.6, SciPy 1.18.0, and pandas 2.3.3. Hyperparameters were fixed before the repeated runs. The checkpoint, configuration, parameter count, training history, software version, device, and runtime are archived for each case–model–seed combination. AdamW uses learning rates $10^{-3}, 10^{-3}, 8 \times 10^{-4}, 6 \times 10^{-4}$ from 30 to 300 buses, weight decay 10^{-5} , and global gradient-norm clipping at 1.0. A 30-epoch cosine schedule has no restart or early stopping. The

Table 1: Test-system characteristics used by the data generator.

Case	Buses	Gen.	Loads	Lines	Trafos	Eligible N-1
IEEE 30	30	6	20	41	0	38
IEEE 57	57	7	42	63	17	62
IEEE 118	118	54	99	173	13	166
IEEE 300	300	69	193	283	128	253

Table 2: Executed pilot training configurations. Width is the latent channel count; K is the number of Chebyshev terms and the polynomial degree is $K - 1$.

Case	Width	Depth	K	Batch	Warm/physics epochs
IEEE 30	64	4	3	16	12/18
IEEE 57	64	4	3	12	12/18
IEEE 118	96	4	3	8	12/18
IEEE 300	96	4	3	4	12/18

adaptive-weight optimizer is Adam with learning rate 0.01. Dropout is 0.05; the directional edge conditioner comprises separate from/to two-layer MLPs of $10 \rightarrow d \rightarrow d$ channels with GELU.

4.2. Contingency and scenario generation

For every in-service line, the generator creates a copied network, sets that line out of service, and tests graph connectivity while retaining transformers and parallel circuits. Islanding outages are recorded and excluded. To maintain a fixed edge axis, topology artifacts preserve the full base-case $\mathcal{B} = \mathcal{L} \cup \mathcal{X}$ canonical branch list, set the selected physical-line status to zero, and retain status one for fixed transformers. Component type and the physical-line-to-branch index map are stored explicitly. Exact Y , Y_f , and Y_t operators are stored once per topology; scenario files reference them by topology ID.

The pilot cap is deterministic and was applied before the split: the base topology and the first 11 non-islanding outages encountered in pandapower line-index order were retained. No random or severity-based outage selection was used. NumPy’s default_rng(2026) then shuffled those 11 outage IDs and the nominal 0.70/0.15/0.15 fractions were floored, yielding 7/1/3 outage IDs plus the base topology in training. This convenience sample does not represent the full eligible N-1 distribution; Table 1 reports the much larger eligible counts.

Loads are perturbed by bounded, spatially correlated multipliers. A squared-exponential kernel is formed from all-pairs graph distances and projected onto a positive-semidefinite eigenspectrum before sampling. Configured ranges are $\pm 30\%$, $\pm 35\%$, $\pm 40\%$, and $\pm 20\%$ for the four pilot systems, respectively. The IEEE 300-bus range was reduced from the initially configured $\pm 50\%$ after repeated IPOPT iteration-limit failures and is therefore a less stressed pilot, not a like-for-like stress test. Active and reactive demand at a load are scaled together to retain its base power factor. Renewable injections are allocated at

generator or load buses and treated as fixed, unity-power-factor injections. All reported experiments use the generator’s correlated synthetic-diurnal profile, with capacity fractions 0.20, 0.20, 0.25, and 0.15, respectively. No claim of validation on real NREL or operational renewable data is made. Specifically, at hour h , the common clear-sky factor is $\max\{0, \sin[\pi(h - 6)/12]\}$, multiplied by an independently sampled Beta(5, 2) cloud factor and site factors $\mathcal{N}(1, 0.08^2)$, then clipped to $[0, 1]$. The graph-kernel correlation lengths for load sampling are 3.0, 3.5, 4.0, and 5.0 hops.

Each feasible scenario is solved with PowerModels’ polar AC formulation and IPOPT [5, 25]. Solver termination status, objective, tolerances, and wall-clock time are retained. The pandapower/PYPOWER backend is available for pipeline debugging but must not be merged into a table labelled “IPOPT.” DC-OPF is solved separately for its speed and approximation baseline using pandapower 3.5.4 runcopp with its default PYPOWER backend and power tolerance 10^{-10} . It minimizes the imported active-power generation cost subject to the linearized active-power/angle model and active branch loading constraints; voltage magnitude and reactive-power variables are absent. No custom angle reference, initialization, or solver option was passed beyond pandapower defaults. Its recorded value is therefore a DC approximation and timing baseline, never an AC-feasibility claim.

Topology IDs, not individual rows, are split once and then held fixed across models and seeds. The realized split is 8/1/3 topology IDs (66.7%/8.3%/25.0%) for each case, with the base topology in training. IEEE 30 has 7/1/3 usable topology IDs because one assigned training outage has no converged sample. The corresponding sample counts are 84/12/36 for IEEE 30 and 96/12/36 for the other cases. Thus, every reported test contingency is absent from training. The exact topology names, outage indices, split assignments, connectivity decisions, and sample counts are published in each case’s topology_manifest.csv; the per-

Table 3: Model-specific fitting and checkpoint-selection protocol. Parameter counts are ordered as IEEE 30/57/118/300. “Best” means the minimum score over all 30 completed epochs; no early-stopping rule was used.

Model	Fitting rows/topologies	Validation rows/topologies	Training objective	Checkpoint score	Parameters
Base-only PINN	10/base only	2/base only	12 supervised + 18 physics	$5\mathcal{L}_{\text{data}} + \mathcal{L}_{\text{pf}} + \mathcal{L}_{\text{th}} + \mathcal{L}_{\text{angle}}$	28 424/45 888/142 712/320 706
Topology-blind GFNO	84 or 96/8 IDs	12/1 held-out ID	12 supervised + 18 physics	same	90 116/90 116/199 684/199 684
Plain GNN	84 or 96/8 IDs	12/1 held-out ID	12 supervised + 18 physics	same	57 348/57 348/125 956/125 956
Data-only GFNO	84 or 96/8 IDs	12/1 held-out ID	30 supervised	same physics-aware score	90 116/90 116/199 684/199 684
GFNO-PINO	84 or 96/8 IDs	12/1 held-out ID	12 supervised + 18 physics	same	90 116/90 116/199 684/199 684

Algorithm 1 Supervised warm start and physics-informed fine-tuning

Require: Topology-disjoint datasets $\mathcal{D}_{\text{tr}}, \mathcal{D}_{\text{val}}$; model $\mathcal{G}_{\Theta}$; warm epochs E_w ; physics epochs E_p

- 1: Initialize Θ , optimizer, scheduler, and adaptive log-weights $\{\log \lambda_m\}$
- 2: for $e = 1$ to E_w do
- 3: for mini-batch $\mathcal{Q} \subset \mathcal{D}_{\text{tr}}$ do
- 4: Predict $\widehat{u}_b = \mathcal{G}_{\Theta}(X_b, E_b, z_b, \mathcal{M}_{\text{pred},b})$
- 5: Update Θ using $\mathcal{L}_{\text{data}}$
- 6: end for
- 7: Evaluate label error and AC residuals on $\mathcal{D}_{\text{val}}$
- 8: end for
- 9: for $e = 1$ to E_p do
- 10: for mini-batch $\mathcal{Q} \subset \mathcal{D}_{\text{tr}}$ do
- 11: Predict $\widehat{u}_b$ with sample-specific topology
- 12: Compute Eqs. (25)–(30) using $Y_b, Y_{f,b}, Y_{t,b}$
- 13: Apply $s_{\text{pf}}(e)$ and $s_{\text{th}}(e)$; activate nonzero objectives and record their first positive baselines
- 14: Update $\{\log \lambda_m\}$ using Eq. (34) with detached targets
- 15: Update Θ using Eq. (31); detach λ_m and clip the global gradient norm
- 16: end for
- 17: Save the checkpoint minimizing $5\mathcal{L}_{\text{data}} + \mathcal{L}_{\text{powerflow}} + \mathcal{L}_{\text{thermal}} + \mathcal{L}_{\text{angle}}$
- 18: end for
- 19: return Best validation checkpoint and all component-wise logs

centages are descriptive, not an unrealized nominal target.

4.3. Baselines

The comparison includes:

1. IPOPT AC-OPF: locally converged dispatch, objective, constraint, and wall-clock reference; no global-optimality certificate is claimed.
2. DC-OPF: linear speed baseline; reactive and voltage feasibility are not attributed to this model.
3. Fixed-topology PINN: a dense PINN trained only on base-topology scenarios and evaluated without re-training on held-out outages.
4. Topology-blind GFNO: a parameter-matched GFNO trained on the same multi-topology rows as GFNO-PINO but supplied one fixed base-topology edge field and Laplacian for every sample.

5. Plain GNN: the same topology inputs and hard output head, with spatial mean aggregation replacing Chebyshev spectral filters.

6. Data-only GFNO: the evaluated graph-spectral architecture trained throughout on solver-label mean-square error, without physics fine-tuning.

The matched topology-blind GFNO is the causal test of topology input. The base-only PINN is retained only as a practically relevant out-of-distribution reference: because it sees 12 base rows rather than the multi-topology training set, its comparison is intentionally not attributed solely to architecture. The plain GNN isolates spectral versus spatial propagation, and the data-only GFNO isolates physics fine-tuning. Width and depth are shared where the architecture permits; parameter counts are archived for every checkpoint.

4.4. Evaluation metrics

For sample b , the signed reference-cost difference is

$$\Delta C_b^{\text{ref}}(\%) = 100 \frac{C(\widehat{P}_b^g) - C(P_b^{g,\text{ref}})}{\max(|C(P_b^{g,\text{ref}})|, C_{\text{scale}})}, \quad (39)$$

$$C_{\text{scale}} = 1 \text{ currency/hour.}$$

A negative value does not establish superiority if the prediction is infeasible. The per-bus complex-power-balance mismatch is

$$d_{i,b} = \sqrt{(r_{i,b}^P)^2 + (r_{i,b}^Q)^2}, \quad d_b^{\max} = \max_{i \in \mathcal{N}} d_{i,b}. \quad (40)$$

Within seed r , “mean bus” is $\bar{d}_{\text{bus},r} = (B_r |\mathcal{N}|)^{-1} \sum_{b=1}^{B_r} \sum_{i \in \mathcal{N}} d_{i,b}$ and “mean max.” is $B_r^{-1} \sum_b d_b^{\max}$. P95 and worst are computed over the $B_r = 36$ held-out samples within each seed. Tables then report the arithmetic mean of each within-seed statistic across five seeds; a displayed $\pm$ value is the sample standard deviation across those five seed statistics. Separate active/reactive maxima use the same nesting. This is a power-balance mismatch in p.u.; it is not labeled a current residual. A telescoping angle-cycle identity is not reported as a performance metric. All reported standard deviations and seed-resampling intervals are computed from only five training seeds per case. They are descriptive summaries of algorithmic variation within this fixed pilot dataset and training budget, not frequentist population-confidence statements.

For constraint class c , the violation rate and magnitude are

$$R_c = \frac{\#\{k : v_{c,k} > \tau_c\}}{\#\{k\}}, \quad (41)$$

$$M_c^{\max} = \max_k v_{c,k}, \quad M_c^{\text{mean}} = \text{mean}_k v_{c,k}, \quad (42)$$

for voltage, thermal, generator, and branch angle-difference constraints. The class threshold is $\tau_c = 10^{-6}$ for every inequality (radians for angle, p.u. otherwise). Within a seed, k ranges over all valid sample–element pairs for rates, while the reported magnitude is the within-seed maximum followed by the five-seed mean. The AC-feasibility proxy requires $d_b^{\max} \leq \tau_{\text{bal}} = 10^{-3}$ p.u. and every represented inequality excess to be at most τ_c . Latency is measured after ten warm-up passes with batch size one and CUDA device synchronization, then summarized across the 36 test samples and five seeds. Neural timing excludes topology construction, host–device transfer, constraint verification, and feasibility restoration; stored solver timing is CPU wall time. Results are grouped by bus count and topology ID. Accuracy on the test partition is the primary unseen-topology result; random row-wise splits are not reported as contingency generalization.

5. Results and discussion

5.1. Executed benchmark and numerical verification

The executed pipeline produced 564 converged AC-OPF labels and 3600 neural test evaluations (4 systems $\times$ 5 models $\times$ 5 seeds $\times$ 36 held-out samples). All AC references were solved with PowerModels 0.21.6, JuMP 1.31.1, and IPOPT 1.15.0 using the polar formulation, optimality and constraint-violation tolerances of 10^{-8} , and the MUMPS linear solver. The released verification suite passes 25 tests. These include an analytic two-bus balance, independent explicit G – B injection equations, directional two-ended branch flows with phase shift, automatic differentiation, hard-box enforcement, angle-limit excess, zero-safe adaptive weighting, topology-disjoint splitting, outage masking, the MATPOWER writer, and numerical agreement with a converged IEEE 30-bus power flow.

Table 4 exposes two important selection effects. First, the IEEE 30 outage line_9_out produced no converged sample in 60 attempts and was retained only as a zero-count training topology; it did not enter the test set. Second, the imported IEEE 57 and IEEE 300 cases required explicit voltage ranges of $[0.8, 1.1]$ and $[0.9, 1.1]$ p.u., respectively, to obtain a runnable pilot. Together with the reduced IEEE 300 load range, these adjustments mean that cross-system comparisons measure computational scaling, not identical electrical stress. Every test set nevertheless contains three contingencies absent from its corresponding training set. The source constraint inventories show MATPOWER angle bounds of $[-360^\circ, 360^\circ]$ on every represented branch, so they are nonrestrictive in

these four cases. Angle excess is still evaluated and included in the proxy so that restrictive source cases cannot be silently omitted.

5.2. Economic performance

Across DC-OPF and the five neural models, Table 5 compares the signed reference-cost difference of Eq. (39); neural entries are five-seed means and standard deviations. A negative value from a neural surrogate is interpreted only jointly with Tables 6 and 7: an infeasible dispatch can appear cheaper by failing to supply losses or demand. The same caution applies to DC-OPF because it omits reactive power, voltage magnitudes, and AC losses. A feasibility-conditioned reference-cost difference is reported only when at least one prediction satisfies the declared proxy; otherwise it is undefined, not zero. Consequently, cost statistics alone do not establish economic superiority over IPOPT. GFNO-PINO’s signed reference-cost differences are $-35.17 \pm 4.15\%$, $-25.32 \pm 1.72\%$, $-26.56 \pm 10.22\%$, and $-20.92 \pm 6.63\%$ from 30 to 300 buses; the corresponding mean absolute differences are 35.17%, 26.44%, 27.85%, and 21.13%. Since no GFNO-PINO sample is proxy-feasible, every feasibility-conditioned difference is undefined.

5.3. Physical residuals and constraint violations

The hard output transformations make voltage and generator box excess zero up to floating-point precision; this is a parameterization check, not learned AC feasibility. Table 6 separately reports thermal and angle-difference excess and the Euclidean active/reactive balance mismatch of Eq. (40). The latter must not be confused with current magnitude or with either component alone. Any nonzero mismatch beyond the declared threshold shows why bounded heads and a small supervised error cannot substitute for a coupled AC-feasibility layer.

Table 9 addresses the logistic-head endpoint limitation. Active reference boxes are common for reactive dispatch in IEEE 57 and 300 and for active dispatch in IEEE 118. The stratified errors do not show one universal direction, but several active groups are materially harder (notably P^s and Q^s in IEEE 30 and P^s in IEEE 57). A projection or hard-sigmoid head could represent endpoints, but would still not guarantee the coupled AC equalities; changing the head would require retraining and is not retrospectively applied to these archived runs.

GFNO-PINO’s mean sample-maximum balance mismatches are 0.281 ± 0.011 , 1.684 ± 0.107 , 0.890 ± 0.112 , and 5.959 ± 3.360 p.u.; five-seed mean within-seed worst values are 0.334, 2.619, 1.121, and 6.351 p.u.

For identical seeds, the paired reductions relative to the data-only GFNO are $74.1 \pm 4.6\%$, $63.4 \pm 2.3\%$, $85.4 \pm 2.0\%$, and $89.3 \pm 6.1\%$. Table 8 shows every paired seed effect and the deterministic seed-resampling intervals $[70.0, 77.2]\%$, $[61.5, 65.1]\%$, $[83.9, 87.0]\%$, and $[83.9, 93.5]\%$. All 20 observed paired effects are positive, but five seeds are too few

Table 4: Executed topology-disjoint pilot dataset. “Failed” counts solver attempts rejected before the target number of converged labels was reached.

Case	Target	Converged	Failed	Usable topol.	Train	Validation	Test	Test topol.
IEEE 30	144	132	154	11	84	12	36	3
IEEE 57	144	144	46	12	96	12	36	3
IEEE 118	144	144	0	12	96	12	36	3
IEEE 300	144	144	0	12	96	12	36	3

Table 5: Signed reference-cost difference (%) on topology-disjoint test sets. Neural entries are mean $\pm$ standard deviation across five training seeds; negative values from infeasible predictions are not optimality gains.

Method	IEEE 30	IEEE 57	IEEE 118	IEEE 300
DC-OPF	-2.65	-3.16	-2.94	-2.03
Base-only PINN (OOD)	1.97 ± 2.66	-7.04 ± 3.74	28.50 ± 1.68	0.45 ± 0.48
Topology-blind GFNO	-34.63 ± 5.03	-22.53 ± 2.74	-24.43 ± 6.28	-22.71 ± 6.32
Plain GNN	-26.78 ± 6.25	-19.08 ± 3.90	-21.02 ± 11.23	-14.33 ± 7.53
Data-only GFNO	3.15 ± 0.37	-3.73 ± 0.90	4.88 ± 0.96	0.67 ± 1.26
GFNO-PINO	-35.17 ± 4.15	-25.32 ± 1.72	-26.56 ± 10.22	-20.92 ± 6.63

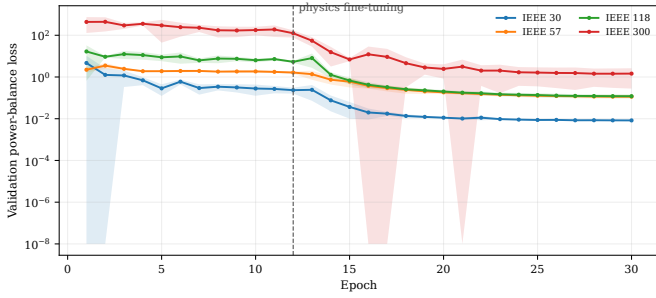


Figure 2: Five-seed mean validation power-balance loss during the 12-epoch supervised warm start and 18-epoch physics fine-tuning. Shading is one standard deviation across training seeds.

to support population-level coverage or robustness claims. The remaining mismatch is two to four orders of magnitude above the 10^{-3} p.u. threshold. GFNO-PINO has a 1.0% thermal-violation rate on IEEE 30 with a five-seed mean maximum excess of 0.00859 p.u. Given the small held-out sample count (36 per seed), this rate should be treated as indicative rather than a precise estimate; per-seed violation counts are archived in the topology-resolved CSVs for independent inspection. The other three systems have no measured thermal excess. Voltage, generator, and the nonrestrictive source angle boxes have zero excess. The resulting feasibility rate is 0% for every learned model and case.

Figures 2 and 3 expose both training variance and the supervised-to-physics transition. Finite weights verify numerical stability of the active-set scheme; they do not imply convergence to the AC equality manifold. The complete per-epoch histories, including raw losses, effective weights, and validation scores, are archived for all 20 GFNO-PINO runs.

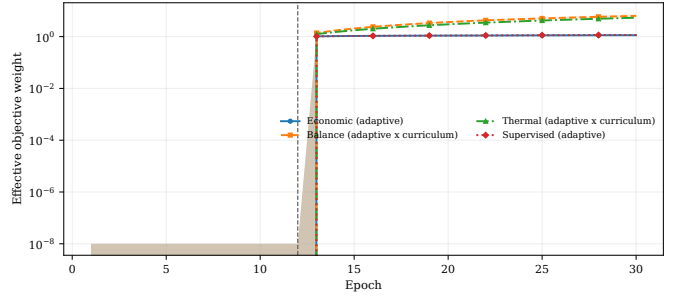


Figure 3: IEEE 30 effective GFNO-PINO objective weights. Curves and shading are the five-seed mean and one standard deviation; balance and thermal curves include their curriculum multipliers. The angle loss remains below the 10^{-8} activation threshold under the non-restrictive source bounds and therefore has zero active-set weight and is not shown.

5.4. Inference latency and scalability

The workstation contains a 13th-generation Intel Core i5-13420H CPU and an NVIDIA GeForce RTX 3050 6 GB Laptop GPU. IPOPT and DC-OPF values are stored CPU solver times; neural values are synchronized float32 CUDA forward-pass times after warm-up. Table 10 is therefore a device-specific comparison, not a hardware-matched solver acceleration. It excludes feature construction, transfer, verification, and restoration. These omissions are material because a prediction that fails the AC-feasibility proxy cannot replace an AC-OPF solve without correction. GFNO-PINO’s five-seed mean CUDA medians are 6.94 ± 0.79 , 5.67 ± 0.25 , 6.19 ± 0.33 , and 10.75 ± 5.93 ms, whereas stored IPOPT CPU medians are 57.29, 204.45, 524.17, and 1484.22 ms. The resulting raw ratios are 8.3, 36.1, 84.7, and 138.1; they compare incomplete surrogate predictions with complete solver outputs, not end-to-end feasible-solution speedups.

Table 6: GFNO-PINO inequality diagnostics on unseen topologies. Each entry is violation rate (%) / mean seed-maximum excess (p.u., except angle in rad). The AC-feasible column additionally requires a 10^{-3} p.u. maximum complex-power-balance mismatch.

System	Voltage	Thermal	Generator	Angle	AC feasible (%)
IEEE 30	0.00/0.00e+00	1.00/8.59e-03	0.00/0.00e+00	0.00/0.00e+00	0.0
IEEE 57	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.0
IEEE 118	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.0
IEEE 300	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.00/0.00e+00	0.0

Table 7: GFNO-PINO complex-power-balance diagnostics in p.u. on unseen topologies. Mean max. is mean $\pm$ seed standard deviation; P95 and worst are five-seed means of within-seed held-out statistics.

System	Mean bus	Mean max.	P95 max.	Worst max.	Mean max. $ r^p $	Mean max. $ r^q $
IEEE 30	0.085	0.281 $\pm$ 0.011	0.324	0.334	0.178	0.232
IEEE 57	0.255	1.684 $\pm$ 0.107	2.415	2.619	1.644	0.843
IEEE 118	0.266	0.890 $\pm$ 0.112	1.020	1.121	0.814	0.668
IEEE 300	0.905	5.959 $\pm$ 3.360	6.221	6.351	5.046	4.696

5.5. Unseen-topology generalization and ablation

Table 11 separates three questions. The matched topology-blind GFNO tests whether the explicit topology field is useful when the training rows are held constant. The plain GNN tests Chebyshev spectral propagation against spatial aggregation. The data-only GFNO tests the effect of physics fine-tuning. Results are averaged first within each case and seed and then across the four cases, so a large system does not dominate by bus count. The base-only PINN is shown as an OOD operational reference but is not used as the causal topology ablation. Five-seed dispersion is supplied in the case-level CSVs; the benchmark supports only effects that remain visible relative to that dispersion.

The observed physics-fine-tuning effect is positive under this pilot protocol, but the architectural conclusions are negative. Averaged across systems, mean sample-maximum mismatch falls from 16.951 p.u. for the data-only GFNO to 2.204 p.u. for GFNO-PINO. In contrast, the matched topology-blind GFNO averages 1.952 p.u. and is better on IEEE 57, 118, and 300; GFNO-PINO is only 0.7% lower on IEEE 30. Thus, these experiments do not demonstrate a causal accuracy benefit from the explicit topology field. The spatial GNN averages 1.790 p.u. and is better than GFNO-PINO on three of four systems, so a Chebyshev advantage is likewise not established. These outcomes do not invalidate the directional topology-conditioned formulation; they show that the present data volume, loss, and training budget are insufficient to validate its claimed architectural advantage. Table 11 includes all declared aggregate physical diagnostics for every learned method. Case-level values, P95 and worst balance mismatch, signed reference-cost differences, label errors, and topology-resolved values remain available in the accompanying CSVs rather than being collapsed into a visually unreadable manuscript table.

6. Limitations

The bounded output layer guarantees independent voltage and generator bounds and the reference angle, but

it does not project onto the full non-convex AC feasible set. Complex-power balance, thermal loading, and branch angle-difference limits are soft objectives; any deployment requires explicit verification and an AC correction solve. Satisfaction of one inequality class cannot repair equality infeasibility. Although the benchmark now uses five seeds, it still uses only 12 selected topology IDs per system, synthetic renewable profiles, and 30 training epochs. It is therefore a controlled outage-generalization study rather than full security-constrained coverage. The deterministic first-eligible selection was not stratified by loading severity, voltage sensitivity, or electrical distance, and severity descriptors for the unselected eligible outages were not generated. Results therefore must not be read as representative of the complete N-1 set. Moreover, the matched topology-blind GFNO and spatial GNN have lower four-system mean maximum balance mismatch than GFNO-PINO. The experiment therefore demonstrates that the implementation can consume unseen topology fields, but not that those fields or the Chebyshev layer improve accuracy under the present protocol.

The method is evaluated only through 300 buses, and the current dense Laplacian batching path is not evidence of scalability to substantially larger interconnections. Although the parameterization is node- and edge-based, the experiments use one checkpoint per grid size; they do not demonstrate one shared checkpoint across 30–300 buses. Generalization is limited to non-islanding single-line outages within a fixed network family and does not cover bus additions, generator-set changes, islands, or simultaneous N-k contingencies. The relaxed IEEE 57/300 voltage ranges and reduced IEEE 300 load range further limit direct cross-case stress comparisons. Finally, IPOPT provides locally converged references rather than global-optimality certificates, and the rejected scenarios—particularly the 154 IEEE 30 failures—may introduce selection bias.

Finally, the localized Chebyshev polynomial is structurally a ChebNet graph convolution. No mesh-refinement, discretization-transfer, or shared mixed-size

Table 8: Observed effect of physics fine-tuning on mean sample-maximum complex-power-balance mismatch (p.u.). Model columns are mean $\pm$ seed standard deviation. Paired reduction compares identical seeds. All five seed effects are shown; the final column is a deterministic 10,000-resample percentile interval used descriptively, not as population-level confidence.

System	Data-only GFNO	GFNO-PINO	Paired mean (%)	Seed effects 2026–2030 (%)	Seed-resampling interval (%)
IEEE 30	1.107 $\pm$ 0.177	0.281 $\pm$ 0.011	74.1 $\pm$ 4.6	77.3, 72.1, 67.0, 77.0, 77.2	[70.0, 77.2]
IEEE 57	4.606 $\pm$ 0.212	1.684 $\pm$ 0.107	63.4 $\pm$ 2.3	65.8, 65.1, 60.2, 63.9, 62.0	[61.5, 65.1]
IEEE 118	6.092 $\pm$ 0.284	0.890 $\pm$ 0.112	85.4 $\pm$ 2.0	82.8, 84.4, 85.0, 88.1, 86.5	[83.9, 87.0]
IEEE 300	55.999 $\pm$ 9.736	5.959 $\pm$ 3.360	89.3 $\pm$ 6.1	93.5, 86.4, 92.9, 93.9, 79.9	[83.9, 93.5]

Table 9: GFNO-PINO error stratified by active reference boxes on the unseen-topology test sets. A reference is “near bound” when it lies within 10^{-4} p.u. of either limit. Fractions use the 36 distinct references per case; active/inactive MAEs pool the five paired model seeds (p.u.).

System	Reference near bound (%): $V/P^s/Q^s$			MAE active/inactive: $V/P^s/Q^s$		
	V	P^s	Q^s	V	P^s	Q^s
IEEE 30	4.81	0.46	6.02	0.022/0.031	0.346/0.092	0.263/0.086
IEEE 57	4.53	21.43	47.62	0.071/0.070	0.620/0.466	0.211/0.394
IEEE 118	7.11	38.68	21.86	0.048/0.042	0.180/0.525	0.198/0.204
IEEE 300	9.80	9.46	40.66	0.098/0.066	0.393/0.682	0.285/0.476

checkpoint experiment was performed. Consequently, this paper uses “GFNO-PINO” only as the archived model identifier and does not claim a neural-operator property beyond topology-conditioned finite-graph prediction.

7. Conclusion

The central finding of this controlled ablation is that physics-informed fine-tuning is the only tested design choice with a consistent improvement under the pilot protocol; benefits from explicit topology conditioning and Chebyshev spectral filtering are not confirmed. The topology-conditioned GFNO-PINO model is therefore an instrument for separating these effects, not a demonstrated superior architecture. Its Chebyshev polynomial filter of the sample-specific normalized Laplacian preserves a graph spectral inductive bias without requiring eigenvector alignment after line outages. Explicit branch status and admittance features make contingencies predictor inputs, while exact Y , Y_f , and Y_l matrices define differentiable AC residuals. Bounded output transformations enforce voltage and generator boxes and the slack reference; they are deliberately not presented as a complete AC-feasibility guarantee.

The executed benchmark uses 564 locally converged IPOPT labels and 3600 held-out neural evaluations spanning four systems, five model variants, and five training seeds. Under this pilot protocol, paired physics fine-tuning runs reduce mean sample-maximum balance mismatch by 63.4–89.3% relative to the data-only GFNO; all 20 observed paired seed effects are positive, while the descriptive resampling intervals are not population-confidence claims. Nevertheless, GFNO-PINO retains 0.281–5.959 p.u. mean maximum mismatch, has a 1.0% thermal-violation rate on IEEE 30, and produces no prediction meeting the declared AC-feasibility proxy. Its four-system balance average (2.204 p.u.) is also worse than the matched topology-blind GFNO (1.952 p.u.) and spatial GNN (1.790 p.u.), so neither a topology-input nor a Chebyshev advantage is

supported. CUDA forward-pass medians yield raw device-specific latency ratios (speedups) of 8.3–138.1 $\times$ relative to stored IPOPT CPU solve medians, but that comparison excludes the correction required for feasibility. Signed negative reference-cost differences are therefore infeasible lower-cost points, not optimality gains.

The defensible role of the present model is a topology-conditioned research prototype for warm starts or screening whose output is independently checked, not a stand-alone certified AC-OPF solver. The negative ablations are scientifically important: representational access to topology is necessary for the intended surrogate, but this benchmark does not show that the network learns to exploit it better than simpler alternatives.

The next step is an explicit differentiable feasibility-restoration layer or post-network Newton/AC-OPF correction, followed by full N-1 coverage, real renewable time series, sparse mixed-size batching, and hardware-matched end-to-end latency accounting. N-k, island-aware operation, uncertainty quantification, and validation on operational networks remain open. Until those steps produce verified feasible outputs, independent solver validation is necessary for safety-critical dispatch.

Data and code availability

The accompanying archive contains the implementation, locked environments, configuration and data-generation files, exact topology and constraint manifests, solver-labelled pilot data, every case–model–seed checkpoint and history, per-sample evaluations, aggregate CSV files, tests, and the scripts that regenerate all tables and figures. A SHA-256 manifest covers the released artifacts, and the README gives a single reproduction command. Source IEEE/MATPOWER cases remain subject to their upstream licenses. The submitted supplement is named `topology_conditioned_gfno_pino_submission.zip`; its companion `topology_conditioned_gfno_pino_`

Table 10: Device-specific batch-one latency in milliseconds. Neural values are five-seed mean $\pm$ standard deviation of medians after warm-up; solver values are stored CPU medians. Preprocessing, transfer, verification, and restoration are excluded.

System	IPOPT AC-OPF	DC-OPF	Topology-blind GFNO	Plain GNN	GFNO-PINO
IEEE 30	57.29	25.52	5.56 ± 0.61	11.38 ± 0.60	6.94 ± 0.79
IEEE 57	204.45	123.24	6.35 ± 0.50	14.23 ± 2.38	5.67 ± 0.25
IEEE 118	524.17	155.79	5.54 ± 0.26	11.08 ± 1.07	6.19 ± 0.33
IEEE 300	1484.22	191.22	9.13 ± 5.32	13.53 ± 0.36	10.75 ± 5.93

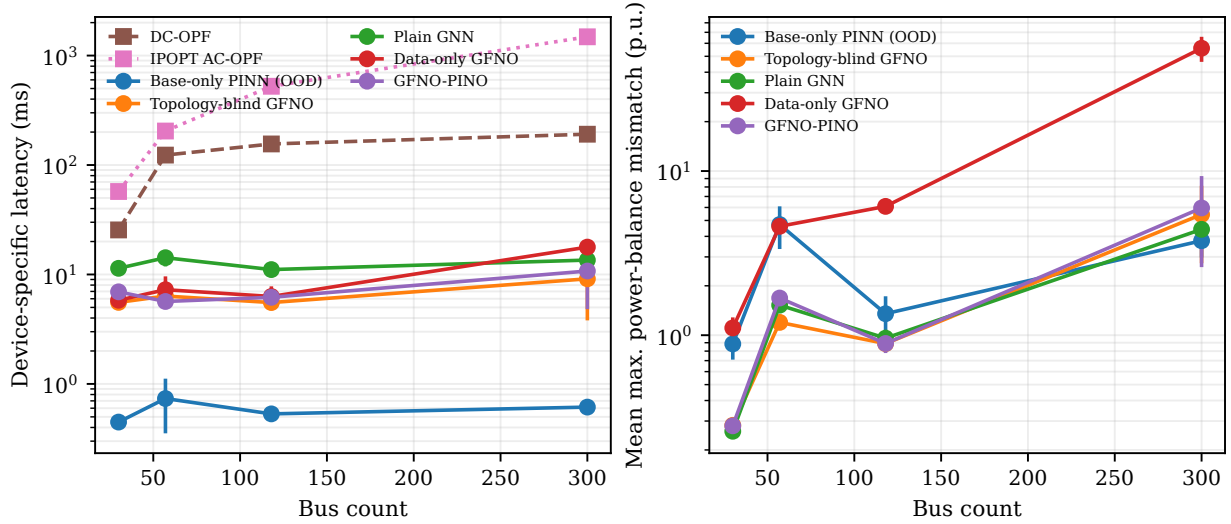


Figure 4: Device-specific median batch-one latency and mean sample-maximum complex-power-balance mismatch versus bus count. Learned-model error bars are one standard deviation across five training seeds. Note: IEEE 57 and IEEE 300 use relaxed voltage bounds and IEEE 300 uses a reduced load-perturbation range relative to IEEE 30/118 (Section 4.2); cross-system points therefore reflect differing electrical stress and should not be read as a controlled scaling sweep.

submission.sha256.json provides the transport checksum. A permanent public repository DOI has not yet been assigned and must be supplied by the authors or journal repository before publication; this manuscript does not invent a locator that was unavailable at revision time.

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Table 11: Four-system, five-seed ablation averages. Mismatch and thermal-excess magnitudes are p.u. Voltage, generator, and angle violation rates and excesses are zero for every method.

Model	Topol. input	Spectral	Physics	$ \Delta C_{ref} $ (%)	Mean bus	Mean max.	Therm. rate (%)	Therm. max.	AC feas. (%)
Base-only PINN (OOD)	Base only	No	Yes	15.0	0.546	2.681	2.98	1.39e-01	0.0
Topology-blind GFNO	No	Yes	Yes	26.8	0.365	1.952	0.25	5.15e-03	0.0
Plain GNN	Yes	No	Yes	21.9	0.372	1.790	0.25	1.74e-03	0.0
Data-only GFNO	Yes	Yes	No	10.9	1.534	16.951	1.63	1.99e-01	0.0
GFNO-PINO	Yes	Yes	Yes	27.6	0.378	2.204	0.25	2.15e-03	0.0

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